ic_autolab round 31 (generic track): report

Protocol: PROTOCOL.md, frozen 2026-11-04T11:14:46Z at git `007707ed2e32b52844a4c24cb025bc9022d2ddb8`; route `local`. Counted GAE only (deterministic, host independent); no wall-time claim; development loop, no validation-gate row. Loop home: `../`; ledger: `../ledger.json`.

Claim record: `PCC-20261104-95313c` (`python3 tools/pcc/pcc.py standing PCC-20261104-95313c`). Until a signed replay and recompute exist, this round's outcome is an unverified assertion [pcc:unverified].

Outcome

k class	sel	sel / inc at w = 0 [95%]	w = 0.1	w = 1	band	S sel (w = 0 / 0.1 / 1)	S inc	S rho-neg	distance sel / rho
20 **null**	wwalkg-x12-n2	0.988 [0.900, 1.085]	0.979 [0.886, 1.081]	0.921 [0.805, 1.054]	sel < inc	2.588 / 2.682 / 3.530	2.620 / 2.741 / 3.831	1.593 / 1.709 / 2.755	1.6x / 1.6x / 1.3x
24 **improvement**	wwalkg-x08-grow-n2	0.788 [0.702, 0.884]	0.776 [0.687, 0.876]	0.707 [0.601, 0.830]	sel < inc	2.164 / 2.243 / 2.955	2.747 / 2.890 / 4.181	1.224 / 1.334 / 2.329	1.8x / 1.7x / 1.3x
28 **improvement**	wwalkg-x08-n2	0.854 [0.758, 0.962]	0.864 [0.763, 0.979]	0.930 [0.798, 1.084]	sel < inc	1.921 / 2.029 / 3.001	2.250 / 2.348 / 3.226	0.932 / 1.019 / 1.800	2.1x / 2.0x / 1.7x
32 **improvement**	wwalk-x10-n2	0.807 [0.708, 0.921]	0.806 [0.698, 0.931]	0.803 [0.659, 0.979]	sel < inc	2.110 / 2.316 / 4.174	2.614 / 2.872 / 5.198	0.872 / 0.958 / 1.727	2.4x / 2.4x / 2.4x

Classes follow the protocol's decision rule. Distance to rho is the distance remaining, not a gate. The reasons per size:

- k = 20: compare ops.status = ok, admissible = true; sel / inc = 0.9879 [0.900, 1.085] at w = 0; band: sel < inc; A/A inc-aa / inc interval [1.000, 1.000] contains 1
- k = 24: compare ops.status = ok, admissible = true; sel / inc = 0.7877 [0.702, 0.884] at w = 0; band: sel < inc; A/A inc-aa / inc interval [1.000, 1.000] contains 1
- k = 28: compare ops.status = ok, admissible = true; sel / inc = 0.8537 [0.758, 0.962] at w = 0; band: sel < inc; A/A inc-aa / inc interval [1.000, 1.000] contains 1
- k = 32: compare ops.status = ok, admissible = true; sel / inc = 0.8072 [0.708, 0.921] at w = 0; band: sel < inc; A/A inc-aa / inc interval [1.000, 1.000] contains 1

Train (development screen, in sample)

k = 20: session `ECBS1h90f95f799984` (run exit 0, verify exit 0, receipt sha256 `6adbcd055d8955939c62adbac44d0694e676347733e0e8c2a53d257690fd61ff`)

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`rho-neg`	rho	- -	`rho.negation`	32	32	1.489 / 1.594 / 2.540	matched reference
`inc`	wsum2g-b15	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=15,free=1`	32	32	2.581 / 2.698 / 3.752	incumbent
`wwalk-x08-n2`	wwalk-x08	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wwalk:x=0.8`	32	32	5.254 / 5.575 / 8.458	exploration: family wwalk-x08 untried at k = 20, N = 0.0001 x N_a
`wwalkg-x08-n2`	wwalkg-x08	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wwalk:x=0.8`	32	32	3.083 / 3.274 / 4.994	exploration: family wwalkg-x08 untried at k = 20, N = 0.0001 x N_a
`wwalk-x08-grow-n2`	wwalk-x08-grow	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wwalk:x=0.8,grow=1`	32	32	4.495 / 4.645 / 5.992	exploration: family wwalk-x08-grow untried at k = 20, N = 0.0001 x N_a
`wwalkg-x08-grow-n2`	wwalkg-x08-grow	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wwalk:x=0.8,grow=1`	32	32	2.628 / 2.742 / 3.767	exploration: family wwalkg-x08-grow untried at k = 20, N = 0.0001 x N_a
`wwalk-x10-n2`	wwalk-x10	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wwalk:x=1.0`	32	32	5.504 / 5.842 / 8.880	exploration: family wwalk-x10 untried at k = 20, N = 0.0001 x N_a
`wwalkg-x10-n2`	wwalkg-x10	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wwalk:x=1.0`	32	32	2.931 / 3.085 / 4.472	exploration: family wwalkg-x10 untried at k = 20, N = 0.0001 x N_a
`wwalk-x10-grow-n2`	wwalk-x10-grow	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wwalk:x=1.0,grow=1`	32	32	5.568 / 5.744 / 7.330	exploration: family wwalk-x10-grow

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
							untried at k = 20, N = 0.0001 x N_a
wwalkg-x10-grow-n2	wwalkg-x10-grow	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.0,grow=1	32	32	3.303 / 3.394 / 4.206	exploration: family wwalkg-x10-grow untried at k = 20, N = 0.0001 x N_a
wwalk-x12-n2	wwalk-x12	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.2	32	32	4.539 / 4.669 / 5.845	exploration: family wwalk-x12 untried at k = 20, N = 0.0001 x N_a
wwalkg-x12-n2	wwalkg-x12	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.2	32	32	2.569 / 2.661 / 3.490	exploration: family wwalkg-x12 untried at k = 20, N = 0.0001 x N_a

k = 24: session `ECBS1h58c486883df3` (run exit 0, verify exit 0, receipt sha256 `97c7fe4bff52353a6dbe9dfb76fab48969e00a2bd257b37cf3d2f12487401a1a`)

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
rho-neg	rho	- -	rho.negation	32	32	1.139 / 1.241 / 2.159	matched reference
inc	wsum2g-b30	2 prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=30,free=1	32	32	2.670 / 2.806 / 4.027	incumbent
wwalk-x08-n2	wwalk-x08	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.8	32	32	3.660 / 4.052 / 7.582	exploration: family wwalk-x08 untried at k = 24, N = 0.0001 x N_a
wwalkg-x08-n2	wwalkg-x08	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.8	32	32	2.763 / 2.947 / 4.599	exploration: family wwalkg-x08 untried at k = 24, N = 0.0001 x N_a
wwalk-x08-grow-n2	wwalk-x08-grow	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.8,grow=1	32	32	2.914 / 3.081 / 4.582	exploration: family wwalk-x08-grow untried at k = 24, N = 0.0001 x N_a
wwalkg-x08-grow-n2	wwalkg-x08-grow	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.8,grow=1	32	32	2.093 / 2.165 / 2.813	exploration: family wwalkg-x08-grow untried at k = 24, N = 0.0001 x N_a
wwalk-x10-n2	wwalk-x10	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.0	32	32	2.657 / 2.819 / 4.277	exploration: family wwalk-x10 untried at k = 24, N = 0.0001 x N_a
wwalkg-x10-n2	wwalkg-x10	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.0	32	32	2.299 / 2.415 / 3.461	exploration: family wwalkg-x10 untried at k = 24, N = 0.0001 x N_a
wwalk-x10-grow-n2	wwalk-x10-grow	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.0,grow=1	32	32	2.659 / 2.821 / 4.278	exploration: family wwalk-x10-grow untried at k = 24, N = 0.0001 x N_a
wwalkg-x10-grow-n2	wwalkg-x10-grow	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.0,grow=1	32	32	2.300 / 2.417 / 3.463	exploration: family wwalkg-x10-grow untried at k = 24, N = 0.0001 x N_a
wwalk-x12-n2	wwalk-x12	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.2	32	32	2.849 / 3.006 / 4.421	exploration: family wwalk-

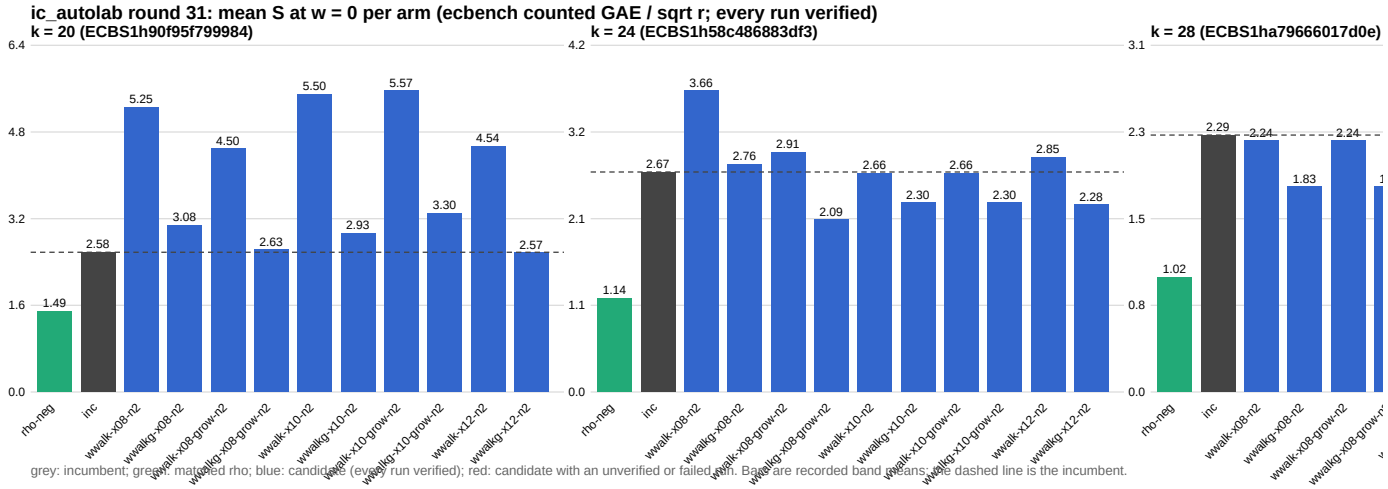
arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
							x12 untried at k = 24, N = 0.0001 x N_a
wwalkg-x12-n2	wwalkg-x12	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.2	32	32	2.276 / 2.369 / 3.202	exploration: family wwalkg-x12 untried at k = 24, N = 0.0001 x N_a

k = 28: session `ECBS1ha79666017d0e` (run exit 0, verify exit 0, receipt sha256 `6ff173fb92d640242e399f6815a16e11cb61253bf1137b2c59c690ed91a5540d`)

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
rho-neg	rho	- -	rho.negation	32	32	1.024 / 1.120 / 1.983	matched reference
inc	wsum2g-b60	2 prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=60,free=1	32	32	2.291 / 2.393 / 3.308	incumbent
wwalk-x08-n2	wwalk-x08	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.8	32	32	2.240 / 2.475 / 4.594	exploration: family wwalk-x08 untried at k = 28, N = 0.0001 x N_a
wwalkg-x08-n2	wwalkg-x08	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.8	32	32	1.831 / 1.930 / 2.820	exploration: family wwalkg-x08 untried at k = 28, N = 0.0001 x N_a
wwalk-x08-grow-n2	wwalk-x08-grow	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.8,grow=1	32	32	2.240 / 2.476 / 4.594	exploration: family wwalk-x08-grow untried at k = 28, N = 0.0001 x N_a
wwalkg-x08-grow-n2	wwalkg-x08-grow	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.8,grow=1	32	32	1.831 / 1.930 / 2.821	exploration: family wwalkg-x08-grow untried at k = 28, N = 0.0001 x N_a
wwalk-x10-n2	wwalk-x10	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.0	32	32	2.219 / 2.411 / 4.140	exploration: family wwalk-x10 untried at k = 28, N = 0.0001 x N_a
wwalkg-x10-n2	wwalkg-x10	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.0	32	32	1.954 / 2.045 / 2.867	exploration: family wwalkg-x10 untried at k = 28, N = 0.0001 x N_a
wwalk-x10-grow-n2	wwalk-x10-grow	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.0,grow=1	32	32	2.219 / 2.411 / 4.140	exploration: family wwalk-x10-grow untried at k = 28, N = 0.0001 x N_a
wwalkg-x10-grow-n2	wwalkg-x10-grow	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.0,grow=1	32	32	1.954 / 2.046 / 2.868	exploration: family wwalkg-x10-grow untried at k = 28, N = 0.0001 x N_a
wwalk-x12-n2	wwalk-x12	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.2	32	32	2.688 / 2.933 / 5.141	exploration: family wwalk-x12 untried at k = 28, N = 0.0001 x N_a
wwalkg-x12-n2	wwalkg-x12	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.2	32	32	2.316 / 2.423 / 3.391	exploration: family wwalkg-x12 untried at k = 28, N = 0.0001 x N_a

k = 32: session `ECBS1he78e7d439059` (run exit 0, verify exit 0, receipt sha256 `b23fb4738d591525aa44a6929e82551205aeee0cc2c5bbf2343b1c46fc4018b8`)

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
rho-neg	rho	- -	rho.negation	32	32	0.856 / 0.940 / 1.696	matched reference
inc	wsum2-b120	2 prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=120,free=1	32	32	2.535 / 2.778 / 4.962	incumbent
wwalk-x08-n2	wwalk-x08	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.8	32	32	2.774 / 3.153 / 6.561	exploration: family wwalk-x08 untried at k = 32, N = 0.0001 x N_a
wwalkg-x08-n2	wwalkg-x08	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.8	32	32	2.836 / 3.038 / 4.859	exploration: family wwalkg-x08 untried at k = 32, N = 0.0001 x N_a
wwalk-x08-grow-n2	wwalk-x08-grow	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.8,grow=1	32	32	2.135 / 2.304 / 3.826	exploration: family wwalk-x08-grow untried at k = 32, N = 0.0001 x N_a
wwalkg-x08-grow-n2	wwalkg-x08-grow	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.8,grow=1	32	32	2.055 / 2.138 / 2.883	exploration: family wwalkg-x08-grow untried at k = 32, N = 0.0001 x N_a
wwalk-x10-n2	wwalk-x10	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.0	32	32	2.051 / 2.246 / 3.998	exploration: family wwalk-x10 untried at k = 32, N = 0.0001 x N_a
wwalkg-x10-n2	wwalkg-x10	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.0	32	32	2.163 / 2.278 / 3.312	exploration: family wwalkg-x10 untried at k = 32, N = 0.0001 x N_a
wwalk-x10-grow-n2	wwalk-x10-grow	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.0,grow=1	32	32	2.051 / 2.246 / 3.998	exploration: family wwalk-x10-grow untried at k = 32, N = 0.0001 x N_a
wwalkg-x10-grow-n2	wwalkg-x10-grow	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.0,grow=1	32	32	2.163 / 2.278 / 3.312	exploration: family wwalkg-x10-grow untried at k = 32, N = 0.0001 x N_a
wwalk-x12-n2	wwalk-x12	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.2	32	32	2.155 / 2.330 / 3.907	exploration: family wwalk-x12 untried at k = 32, N = 0.0001 x N_a
wwalkg-x12-n2	wwalkg-x12	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.2	32	32	2.276 / 2.382 / 3.336	exploration: family wwalkg-x12 untried at k = 32, N = 0.0001 x N_a



grey: incumbent; green: matched rho; blue: candidate (every run verified); red: candidate with an unverified or failed run. Bars are recorded band means; the dashed line is the incumbent.

train screen

Selection

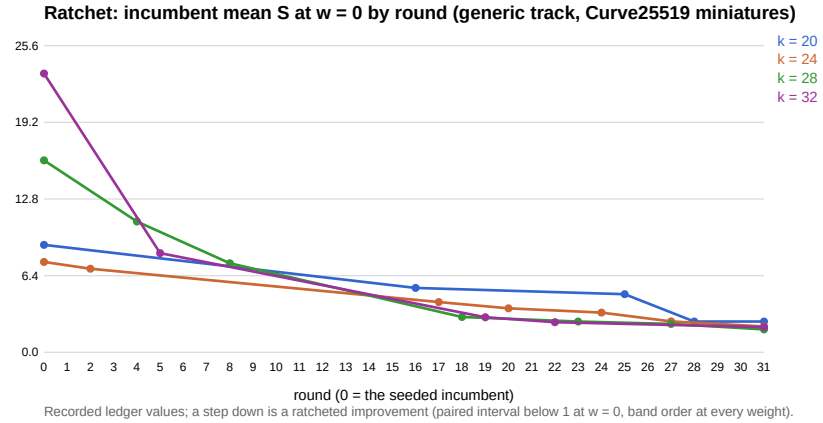
- k = 20: challenger `wwalkg-x12-n2` (Pareto alternative at w = 1: `wwalkg-x12-n2`)
- k = 24: challenger `wwalkg-x08-grow-n2` (Pareto alternative at w = 1: `wwalkg-x08-grow-n2`)
- k = 28: challenger `wwalkg-x08-n2` (Pareto alternative at w = 1: `wwalkg-x08-n2`)
- k = 32: challenger `wwalk-x10-n2` (Pareto alternative at w = 1: `wwalkg-x08-grow-n2`)

Holdout sessions

k	session	run exit	verify exit	receipt sha256	levels
20	`ECBS1h95ee8f44e876`	0	0	`e351712b0986dcfc5375be305d1454a5e21c65b687d2377df323949dea4a0c37`	{“verified”:480}
24	`ECBS1h74e92e8a400c`	0	0	`6f94145bfcc2e1fe79e609866b1cf4942d8b3d51f389d29687786d19eb7613a4`	{“verified”:480}
28	`ECBS1h9970d0e17516`	0	0	`1d0d7c8e529cea8d7d26b263e8049b44779c10ba7840fa3309bec2bf0f69626f`	{“verified”:480}
32	`ECBS1hedd879993025`	0	0	`0bcb9ac6dcafcac392ee7e88bb0d8a428bed45746e66f5813978aae783f90299`	{“verified”:480}

Ledger after this round

k	incumbent	since round	S (w = 0 / 0.1 / 1)	distance to rho
20	`wsum2g-b15-n2` (`prime-projected-abscissa:size=2,scan_cap=32` + `wsum:coef=15,free=1`)	28	2.575 / 2.691 / 3.741	1.8x / 1.7x / 1.5x
24	`wwalkg-x08-grow-n2` (`prime-projected-abscissa:size=2,scan_cap=32` + `wwalk:x=0.8,grow=1`)	31	2.164 / 2.243 / 2.955	1.8x / 1.7x / 1.3x
28	`wwalkg-x08-n2` (`prime-projected-abscissa:size=2,scan_cap=32` + `wwalk:x=0.8`)	31	1.921 / 2.029 / 3.001	2.1x / 2.0x / 1.7x
32	`wwalk-x10-n2` (`prime-projected-abscissa:size=2,scan_cap=32` + `wwalk:x=1.0`)	31	2.110 / 2.316 / 4.174	2.4x / 2.4x / 2.4x



Recorded ledger values; a step down is a ratcheted improvement (paired interval below 1 at w = 0, band order at every weight).

ratchet

Evidence and deviations

- ecbench binary sha256: `9f3a25b151d9868419ec86574451424a506f06c9bf8e5d790b44d48428feb4eb`; `--cpus auto`, allow_busy = true; `verify --replay 8` per session; audit receipts in `analysis/`; sessions in `sessions/` (records in the artifact store when large).
- Route: the local route is an explicitly requested development screen, not the Iso Lab EKS default; counts are deterministic and replayed, wall time is not claimed. No row of this round enters the IC leaderboard or `benchmarks/README.md` (validation gate).
- Train selection is in sample (disclosed); the holdout is fresh (new `target_seed`, excluded exposures). Failed, timed-out and regressing arms are retained above.